

# VIKTOR LAURENS DE GROOTE

Master of Science in Electromechanical Engineering: Robotics

## LANGUAGES

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|---|---|
| **Dutch** | Native |
| **English** | C1 |
| **French** | B2 |
| **German** | B1 |
| **Norwegian** | A1 |

## HARD SKILLS

**Programming**
Python • MATLAB / Simulink
C++ • Java

**Machine Learning**
PyTorch • NumPy / SciPy
OpenCV

**Control & Optimization**
MPT3 • Control System Toolbox
GTSAM • Ceres Solver
CasADi (basic)

**Robotics & Simulation**
ROS 2 • PyBullet
Robotics Toolbox
MoveIt 2
Isaac Sim / Isaac Lab (basic)

**3D Reconstruction & Rendering**
COLMAP • Metashape (basic)
3D Gaussian Splatting • Unity

**Development Tools**
Linux • Git / GitHub

**CAD & Engineering**
Autodesk Inventor

**Productivity**
MS Office

## SOFT SKILLS

Teamplayer • Leadership
Taking initiative • Result-oriented
Social • Creative • Patient
Analytical • Problem-solving
Strong listening

## CONTACT


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SCAN ME

## PROFILE

My fascination with dynamical systems began at 16, when a mathematics project on chaos theory introduced me to the Lorenz attractor. I was intrigued by how a few deterministic nonlinear equations could produce such complex and unpredictable behaviour. Years later, I discovered control theory where we move from studying what a system will do to investigating what we can make it do, and under which conditions we can guarantee the outcome. I am drawn to research because it challenges us to question what we think we know, investigate what we cannot yet explain, and develop knowledge that extends what is possible. What drives me is the desire to understand challenging problems deeply, turn that understanding into working technology, and contribute to scientific progress that makes a meaningful difference in the world around us.

## EDUCATION

**Master of Science Electromechanical Engineering: Robotics** 2022 - 2024
**Vrije Universiteit Brussel (VUB) & Université Libre de Bruxelles (ULB)**

Graduated Magna Cum Laude with control-oriented electives including Advanced Control (19/20), and Optimization-based Control Design (17/20). Projects included control for knuckle-crane load-swing suppression; PINN for pendulum prediction and parameter identification; vibration-isolation modelling for Einstein Telescope; development of own finite-element analysis software; PLC programming; real-time embedded systems (RTOS); and model predictive control for mobile robots. Master thesis: developed and benchmarked an algorithm combining Conflict-Based Search (CBS) and Probabilistic Roadmaps (PRM) for motion planning of manipulators.

**Bachelor of Science Engineering: Electromechanical Engineering** 2019 - 2022
**Vrije Universiteit Brussel**

Graduated Magna Cum Laude. I was assigned the role of student delegate and during these years. In collaboration with ArcelorMittal Belgium and other students, I prepared a course text for the company around the drive of DC motors.But the most memorable project was building a rope-climbing robot.

# HOBBIES

Tennis • Padel • Golf • Travel
Cycling • Running • Cooking
Engineering projects • Chess

# EXTRA-CURRICULAR

**Sept - Dec 2022**

**Driverless teammember**
VUB Racing

Formula SAE is an international student design competition organised by SAE International. Starting that year VUB Racing also build a driverless vehicle to compete in the Formula Student Driverless section.

**Sept 2020 - June 2021**

**Student representative**
Student council

I served as a vital link between the student body and faculty/administration, advocating for student interests and welfare. Actively participated in council meetings, contributed to the development and implementation of policies affecting academic and campus life, and organized feedback sessions to gather student opinions and concerns.

# WORK EXPERIENCE

**ORMIT Talent Trainee** — Mar 2025 - July 2026
**Buildwise (Belgian Construction Research Institute) - Brussels, Belgium**

Conducted applied research at Buildwise, Belgium's innovation centre for the construction sector, focusing on learning-based visual perception and 3D reconstruction for building assessment and construction monitoring. Trained and fine-tuned pretrained vision models using PyTorch for semantic and instance segmentation, including dataset preparation, model evaluation, and benchmarking. Developed 3D reconstruction pipelines combining EKF-based visual-inertial SLAM for online state estimation with offline global bundle adjustment and factor-graph optimization using GTSAM and Ceres Solver, enabling real-time visualization of scanning coverage and improved reconstruction accuracy. Contributed to software applications that translated these methods into practical tools for building-condition assessment, renovation planning, and construction progress monitoring through comparison with BIM models.

**Internship at Cybernetics Department** — July - Oct, 2023
**Norwegian University of Science and Technology (NTNU) - Trondheim, Norway**

Engaged in research around distributed communication-aware path planning for autonomous underwater fleets. My contribution was an optical communication framework for autonomous underwater vehicles using Hydromea Luma optical modems.

**Volunteer Security Teamleader** — August, 2023
**PSTEREO Music Festival in Trondheim, Norway**

Ensuring attendee safety through monitoring grounds, managing crowds and swiftly addressing incidents. Enhancing joy of festival experience through direct engagement with attendees and providing them with information and assistance, fostering a welcoming and supportive atmosphere.

# CERTIFICATIONS

Directly after my studies, I further expanded my knowledge through online courses by UC Berkeley and MIT focused on machine learning (CV, NLP, decision-making), reflecting my commitment to continuous learning and growth.

**Artificial Intelligence: Business Strategies and Applications** — Nov - Feb 2025
**Berkeley Haas School of Business- Berkeley (California), USA**

**Applied Data Science Program: Leveraging AI for Effective Decision-Making** — Nov - Apr 2025
**MIT Professional Education - Cambridge (Massachusetts), USA**